1. If the standard deviation of a dataset is small, what does this tell you about the data points?

(A) They are widely dispersed.
(B) They are clustered closely around the mean.
(C) The data is skewed.
(D) The data is symmetrical.

2. Consider the small data set from the notes.

6 7 7 7 8 8 9 9 10 11 11

If every value in this data set is increased by 10, what happens to:

- the mean and median of the set?

(A) Both remain the same.
(B) Both increase by 10.
(C) The mean increases by 10, but the median remains the same.
(D) The mean remains the same, but the median increases by 10.

- the variance and standard deviation of the set?

(A) Both remain the same.
(B) Both increase by 10.
(C) The variance increases by 10, but the standard deviation remains the same.
(D) The standard deviation remains the same, but the variance increases by 10.

3. For which one of the following distributions will the median be a better measure of center than the mean?

(A) Salary data for players in the National Basketball Association (NBA) where most of the players earn the league minimum and a few superstars earn very high salaries in comparison.
(B) Repeated weight measurements of the same 1.6-ounce bag of Peanut M&Ms by students in a large chemistry class using a non-digital balance scale.
(C) Height data from a large random sample of men.
(D) Exam scores with a central peak around an average test score and a few students scoring lower than the average and a few scoring higher.

4. Write down the correct `dplyr` function for each task to be performed on a data frame.

- I want to calculate a summary statistic(s) on a variable in my data frame. _______________
- I want to isolate a variable(s) of interest, so I make a new data frame from my original data frame with the same number of rows but a smaller number of columns. _______________
- I want to modify the properties of an existing variable in my data frame (or make a new variable from existing variables in my data frame) _______________
- I want to use `summarise()` to calculate a summary statistic(s) on a variable in my data frame, but I would like this calculation to happen separately among different sets of observations in the variable. _______________

5. Sketch your best sense of the distribution of the following variable(s). For each, please:

i. Use a form of statistical graphic that emphasizes the important elements of the distribution.
ii. Label the axes and provide plausible values for the tick marks.
iii. Describe in words the shape of the distribution.
iv. State which measure of center and spread would be most appropriate and approximate their values.

Make a note of any assumptions you're making in interpreting these variable names.

How do Berkeley students get to campus?

Scores on a quiz where most do well but a few score poorly